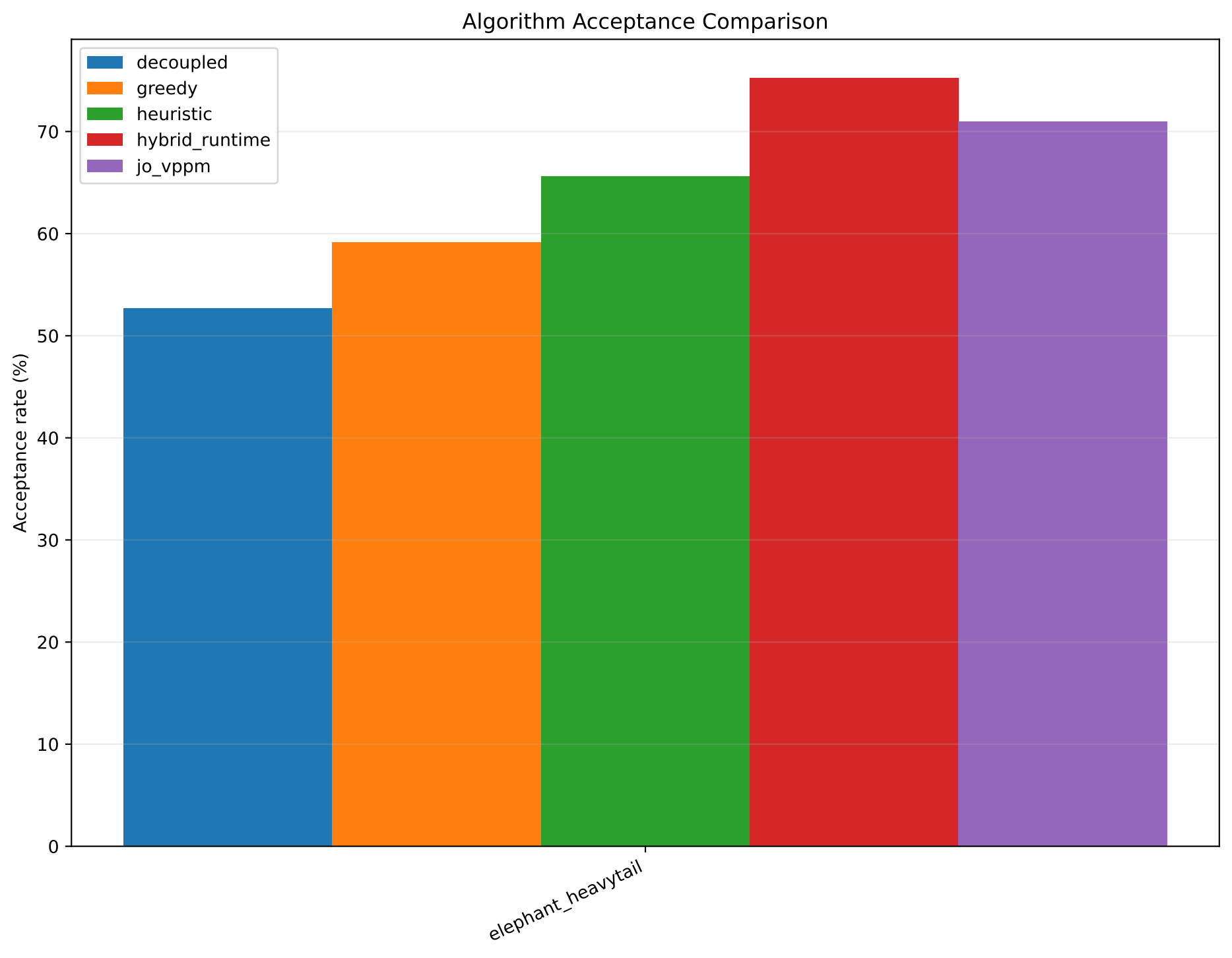
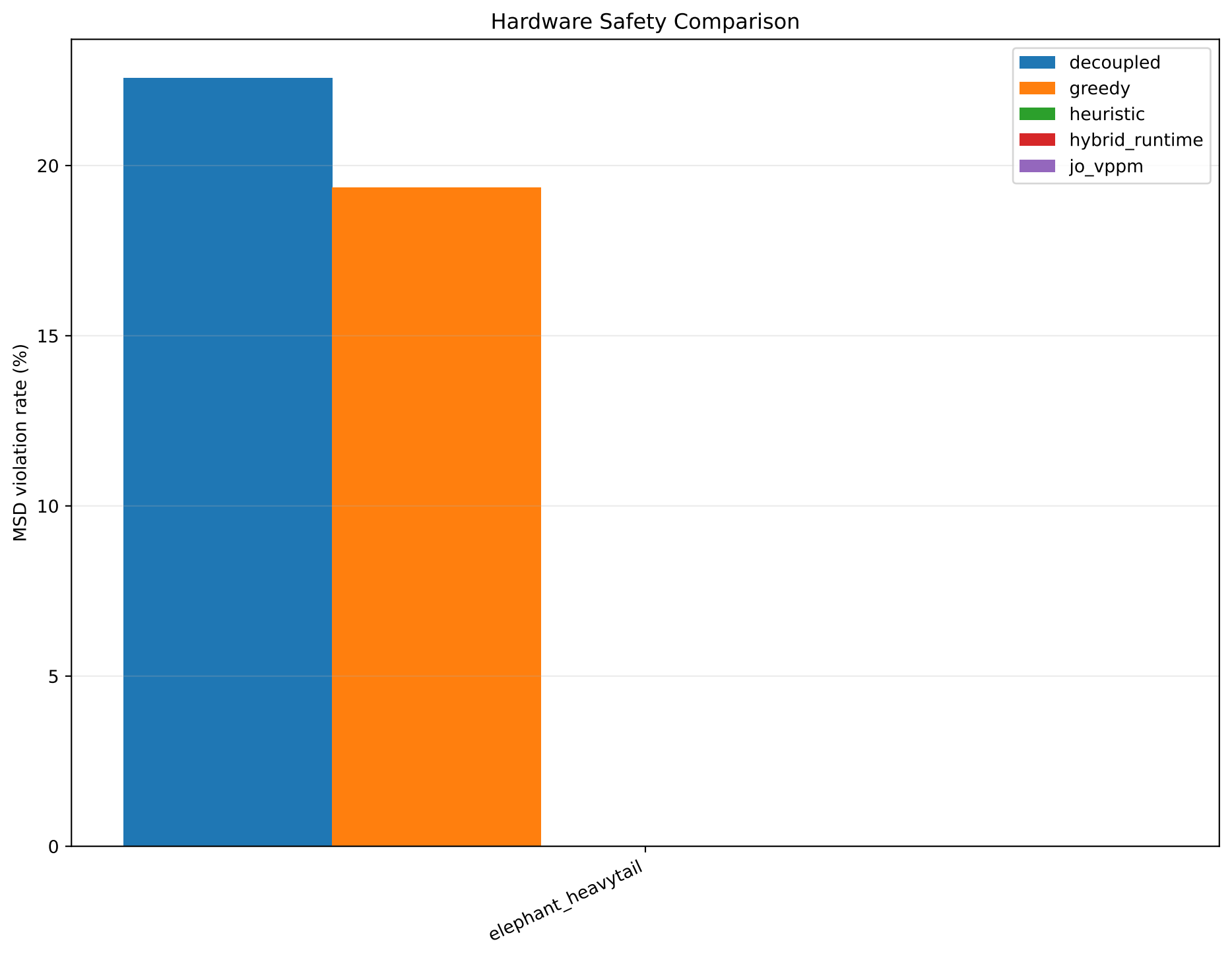
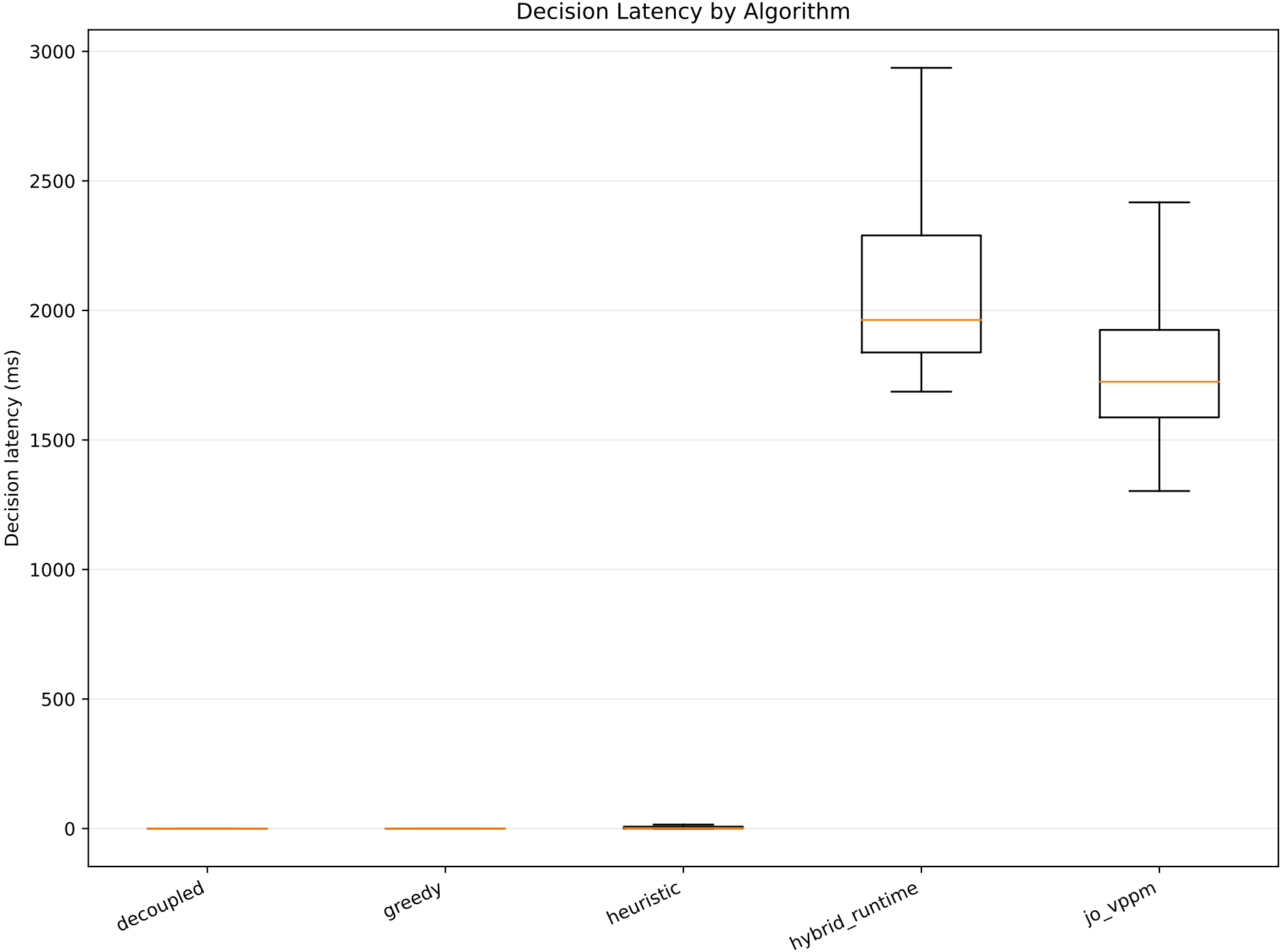
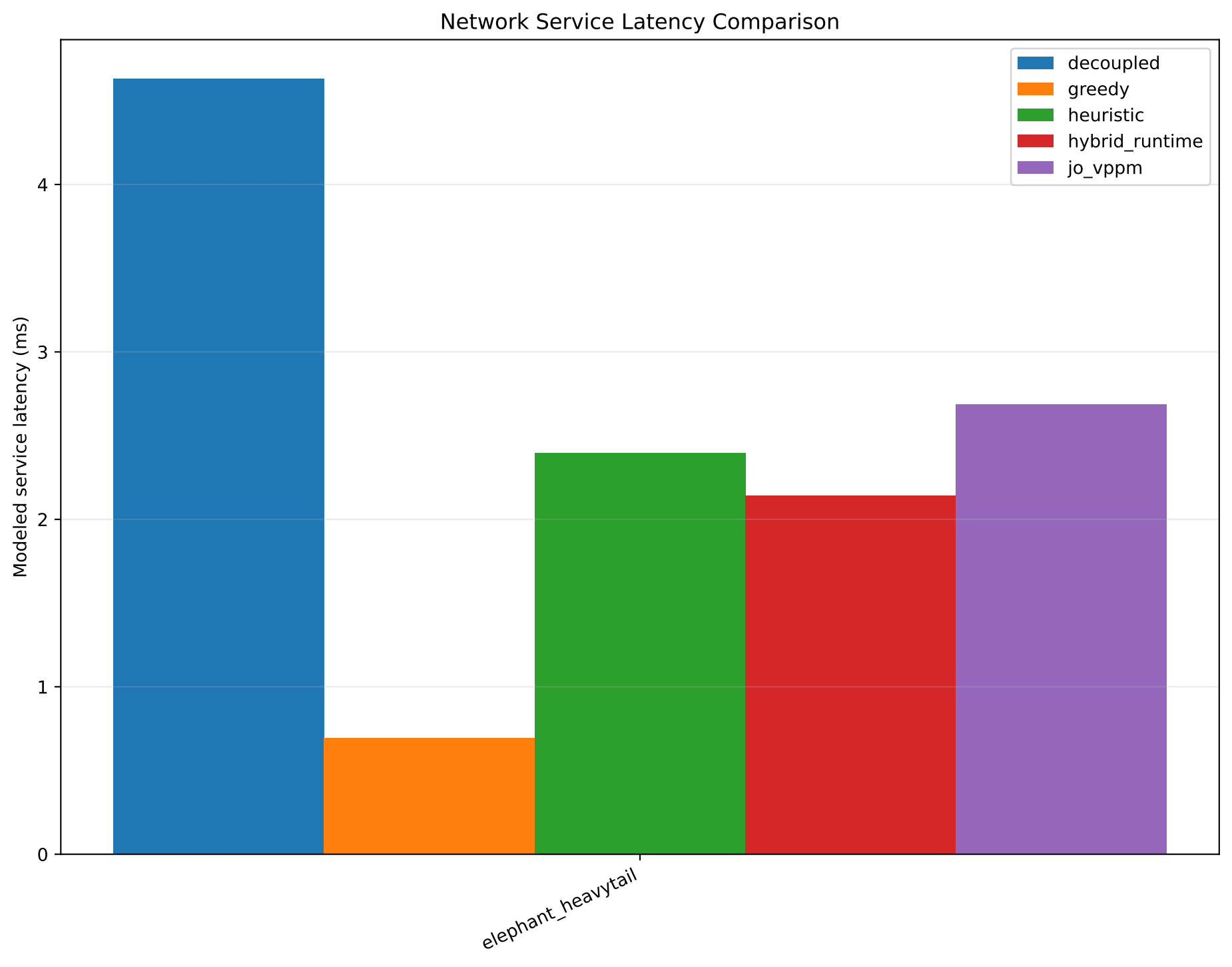
3S-COM Testbed Scenario Report - Scenario 2 - Elephant Heavy-Tail

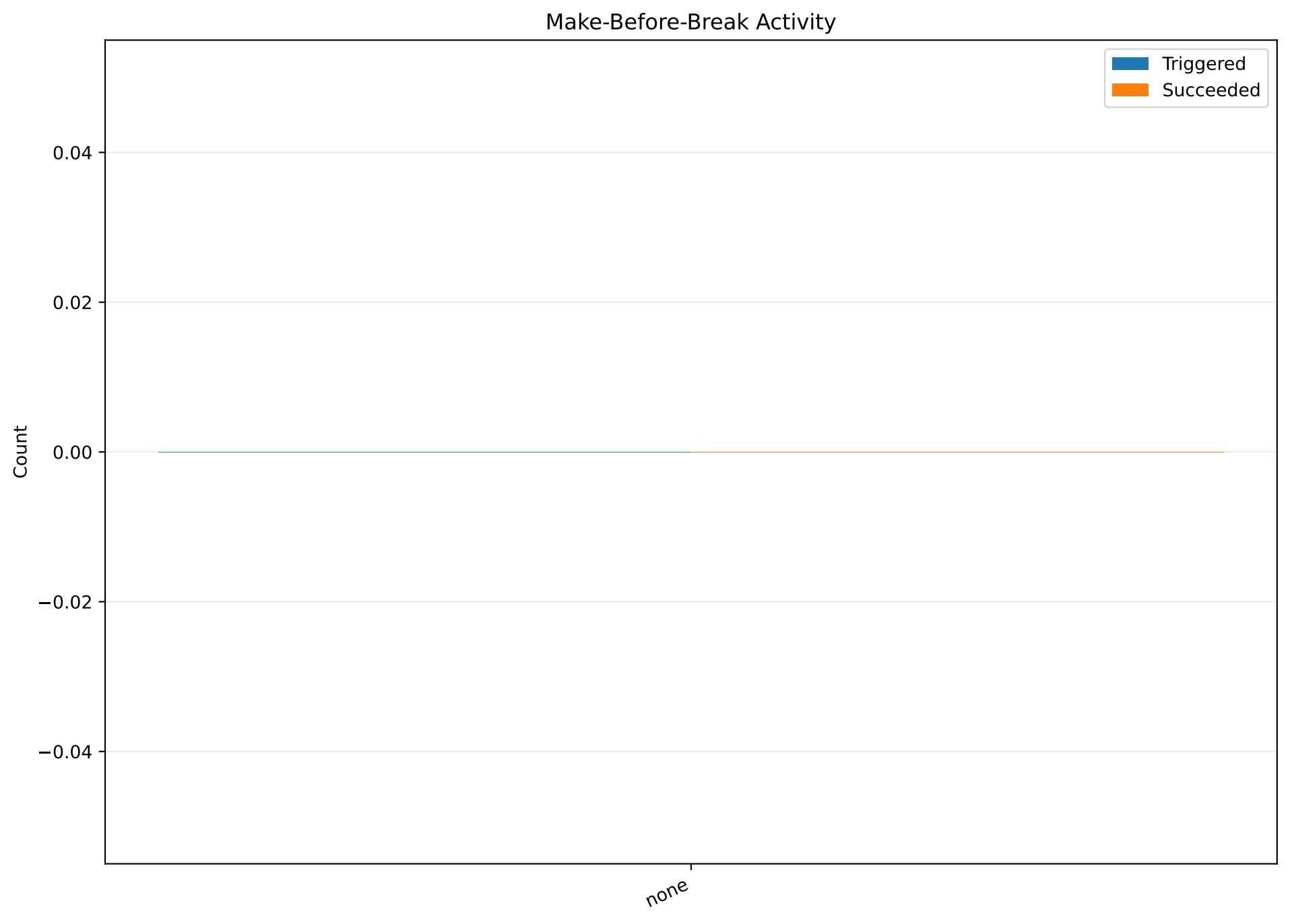
Scenario	Algorithm	Requests	Accept	No safe	MSD viol	Decision ms	Service ms	Timeouts	Migrations	MSD drops
io 2 - Elephant H	ybrid_runtime	93	75.3%	24.7%	0.0%	2011.4	2.14	0	0	0
io 2 - Elephant H	jo_vppm	93	71.0%	29.0%	0.0%	1823.9	2.69	0	0	0
io 2 - Elephant H	heuristic	93	65.6%	34.4%	0.0%	5.9	2.40	0	0	0
io 2 - Elephant H	greedy	93	59.1%	21.5%	19.4%	0.6	0.70	0	0	0
io 2 - Elephant H	decoupled	93	52.7%	0.0%	22.6%	0.4	4.63	0	0	0


Algorithm Acceptance Comparison
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
Acceptance rate (%)
0
10
20
30
40
50
60
70
elephant_heavytail


Hardware Safety Comparison
MSD violation rate (%)
0
5
10
15
20
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
elephant_heavytail


Decision Latency by Algorithm
Decision latency (ms)
3000
2500
2000
1500
1000
500
0
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm


Network Service Latency Comparison
Modeled service latency (ms)
0
1
2
3
4
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
elephant_heavytail


Make-Before-Break Activity
Triggered
Succeeded
0.04
0.02
0.00
−0.02
−0.04
Count
none